$\longleftrightarrow = \rightleftarrows$

Date
DELTA Pg No.

4.1 [Neio] ⇒ Travelling Salesman Problem

⇒ Given a directed weighted graph
⇒ Every edge is having wt, so a cost adjacency matrix is shown along.

Problem :→ We have to start from some vertex and travel all the other vertices one and have to return to the same starting vertex. The cost of travelling should be minimum.

Formula ⇒ $g(i, S) = \min_{K \in S} \{ c_{iK} + g(K, S - \{K\}) \}$

gives cost of a route

recursive formula.

↓

cost of route is defined in terms of cost of the route again

general formula.

cost of min^m cost path visiting each ~~also~~ city in set S exactly once, starting at 1 and ending at i

Cost fcn for each node ⇒
i ⇒ starting index/node

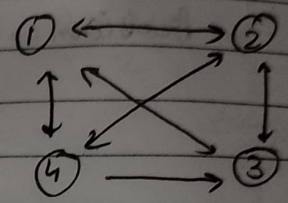
	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

have to visit these

$g(1, \{2,3,4\})$ — cost of going 1 → K

$$= \min_{K \in \{2,3,4\}} \{ C_{1K} + g(K, \{2,3,4\} - \{K\}) \}$$

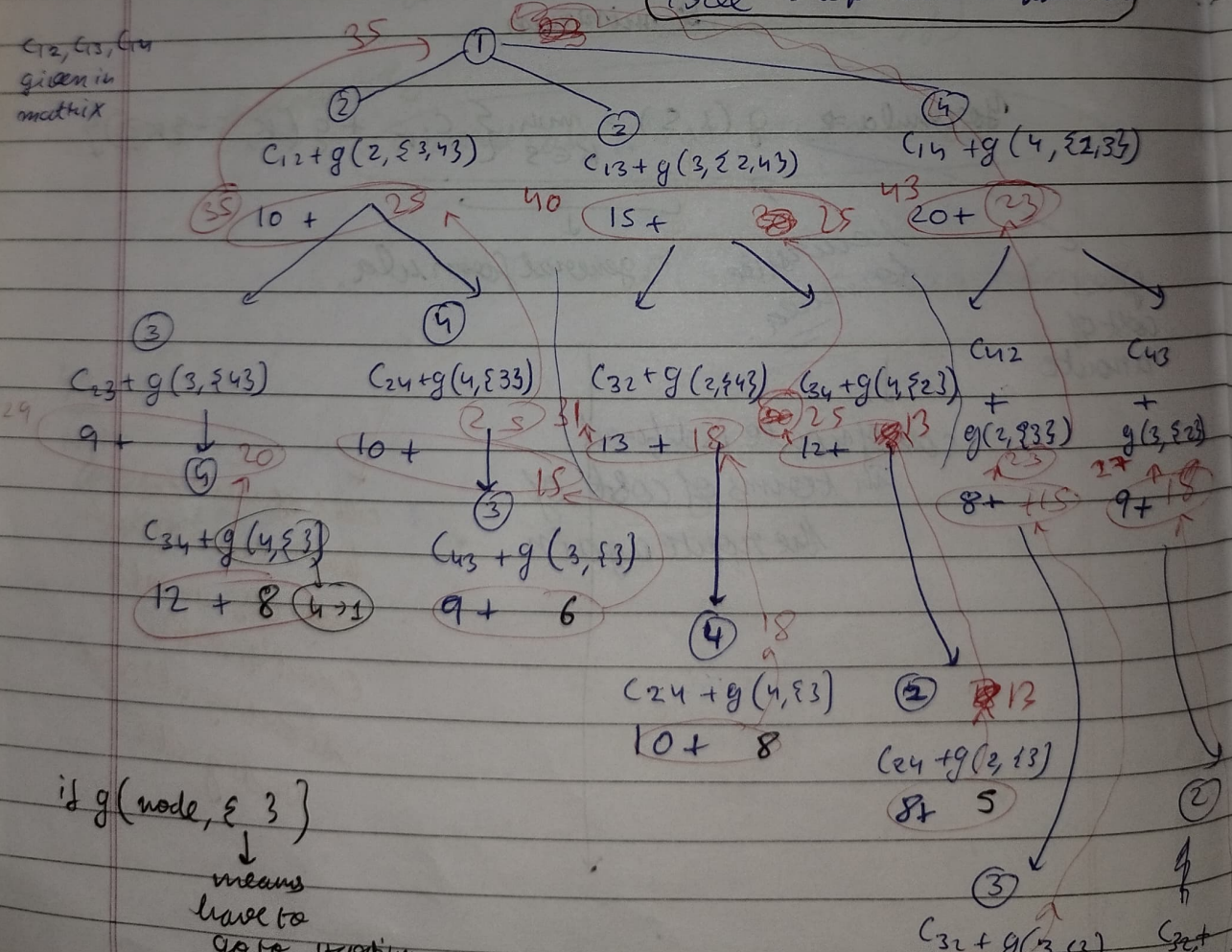
cost of going from K to the rest.



since recursive, let's generate a recursive tree.

Tree → top → down approach

C_{12}, C_{13}, C_{14} given in matrix



if $g(\text{node}, \{3\})$ → means have to go to starting node.

Although ↓ should do recursively ↓ but need the bottom most/base values first, so for convenience ↓ bottom up approach

↓ used tree to move up

So, maybe have to first make tree, then table to see more up)

Although ↓ should do recursively ↓ but need the bottom most/ base values first, so for convenience ↓ we follow bottom up approach.

(Six) → Dynamic Prog ⇒ Table analysis
Dynamically generated.

We don't generally use recursive tree, to easily calc, we make it as a table in bottom-up approach.

$g(2, \phi) = 5$
$g(3, \phi) = 6$
$g(4, \phi) = 8$

$g(2, \{3\}) = C_{23} + g(3, \phi) = 9 + 6 = 15$
$g(2, \{4\}) = C_{24} + g(4, \phi) = 10 + 8 \Rightarrow 18$

$g(3, \{2\}) = C_{32} + g(2, \phi) = 13 + 5 \Rightarrow 18$
$g(3, \{4\}) = C_{34} + g(4, \phi) = 12 + 8 = 20$

$g(4, \{2\}) = 13$
$g(4, \{3\}) = 15$

(Ex) ↓ used tree to move up

So, maybe have to first make tree, then table so easily move up)

use tree →

$g(2, \{3, 4\}) = \min\left(C_{23} + g(3, \{4\}) + (C_{24} + g(4, \{3\})\right)$

29 ↑ 25 ✓

$g(3, \{2, 4\}) = 25$

$g(4, \{2, 3\}) = 23$

$g(1, \{2, 3, 4\}) \Rightarrow$ min of $(35, 40, 43)$

$10 + 25, 15 + 25, 20 + 23$

$C_{43} + g(3, \{2\}) + 15$

$C_{32} + g(2, \{3\})$
$13 + 5$